

1 Project Execution Roadmap

This roadmap turns the proposed study into team assignments and a reusable Rust tool. The research question and graph-level Hodge method remain those of the proposal. Detailed tasks, dependencies, file contracts, and acceptance checks are recorded in `ROADMAP.md`. Environmental-response evidence prioritizes genes; without patient exposure measurements, it does not establish that an exposure caused a splicing change.

1.1 Team Workstreams

The genetics team will collect susceptibility evidence, normalize stable gene identifiers, retain source provenance, and export a checked gene table. The environment team will record exposure and tissue context, build the corresponding response-gene table, and prepare risk, environmental, and overlap memberships. These two tasks can proceed in parallel.

The data team will obtain the selected public endometriosis and control data, verify tissue and sample metadata, obtain or derive junction counts, and export quality-checked standard tables. The mathematics team will specify graph orientation, incidence matrices, normalization, projections, numerical tolerances, and synthetic expected answers. Dataset preparation and synthetic validation can proceed simultaneously.

The Rust team will build input validation, sample and gene grouping, graph and matrix construction, decomposition, comparison, ranking, and export modules. Parser development can begin with synthetic inputs before real data are ready. The statistics team will compare per-sample measurements, report effect sizes, validate permutation tests, correct for multiple testing, and compare simpler baselines. The presentation team will prepare pipeline and graph figures, reviewed biological explanations, and a small live demonstration. Team members may share roles, but each deliverable will have an owner and a reviewer.

1.2 Development Phases

The phase numbers identify dependencies rather than requiring every task to run in sequence. Schemas and mathematical checks must be agreed before real-data integration; biological interpretation follows validated statistical outputs.

Phase	Main work	Deliverable
1	Organize sources, roles, and interfaces	Assigned tasks and data contract
2	Build gene evidence and memberships	Risk, environment, candidate tables
3	Prepare data and verify metadata	Standard junction tables and QC
4	Validate synthetic mathematics	Known answers and numerical checks
5	Implement Rust core	Synthetic end-to-end CLI
6	Integrate real junction data	Checked real-data run
7	Compare disease and control	Per-sample values and differences
8	Test effects and compare baselines	Statistical and baseline outputs
9	Interpret supported findings	Evidence-linked explanations
10	Complete tool usability	Documented, tested Rust release
11	Rehearse small live example	Fast demo and offline backup
12	Assemble report and presentation	Figures, slides, reproducible methods
13	Optional proposal extensions	Separately validated extensions

Core phases are required. Performance improvements are desirable when needed; patient-level exposure analysis, full simplicial Hodge analysis, genome-wide scaling, and other-disease applications remain optional extensions after the core deliverables are ready.

1.3 Minimum Viable Tool

The minimum software deliverable is a reusable Rust command-line program. It will read standardized junction counts, validate sample and gene identifiers, construct directed splice graphs and the incidence matrix, normalize junction usage, and calculate the gradient and cycle-space projections defined in the proposal. It will retain per-sample measurements, compare eligible case and control groups, rank genes, identify contributing junctions, and export both readable summaries and machine-readable results with run parameters.

Synthetic validation will cover linear and branching trees, exon skipping, back-splicing, noise, and changing junction weights. Checks will verify signal reconstruction, cycle-space membership, orthogonality, and energy accounting. A graph cycle-space component does not by itself establish a directed biological cycle or a circular RNA molecule. Zero-support inputs and numerical failures must receive explicit statuses.

The following commands illustrate the intended interface; they are not yet implemented, and the tool name is a placeholder:

```
tool analyze sample.tsv
tool gene GENE_ID --case cases/ --control controls/
tool compare --case cases/ --control controls/
tool rank --genes candidate_genes.tsv \
  --case cases/ --control controls/
```

The first release may analyze a small candidate set. It must remain usable with new conforming input files and include installation instructions, input and output specifications, example data, numerical tests, and actionable errors. A report or notebook alone does not satisfy the software deliverable.

1.4 Midpoint Milestone

By midpoint, the team should have documented gene-set criteria and provenance, candidate tables, a selected accessible dataset with reviewed metadata and a prepared small junction subset, and stable input/output schemas. Synthetic expected answers and mathematical validation should be ready. The Rust parser, graph constructor, and incidence-matrix code should work, with one synthetic example running through normalization, decomposition, and export. Final biological results are not required at this checkpoint. Any missing data prerequisite must remain recorded as an outstanding task.

1.5 Final Deliverables

The final deliverables are a documented reusable Rust CLI, integrated real endometriosis/control data, validated per-sample measurements, group effects, statistical testing where the sample design permits it, baseline comparisons, and evidence-linked interpretation. Statistical outputs will retain sample counts, exclusions, permutation settings, and multiple-testing adjustments. Figures and final documentation will distinguish observed findings from illustrative examples and state the limitations of the analysis.

The live demonstration will show a tiny synthetic graph, run its decomposition, and then run one reviewed real gene through the case/control command. The team will explain the nodes, junctions, cycle-space measurement, group difference, and contributing junctions. Each live command should finish in seconds during rehearsal. Larger analyses will be precomputed and clearly labeled. Saved output, matching figures, and a short recording will provide an offline backup.

1.6 Risks and Fallbacks

If the overlap gene set is small, the team can retain the risk and environmental sets separately, as already allowed in the proposal. If junction tables are unavailable, the data team can derive them from available sequencing files with established software. Poor coverage or unsuitable meta-data must be documented; insufficient biological replication supports descriptive comparisons, with inference explicitly unavailable.

If the full dataset is too large, the team will use the proposal’s small-gene-set MVP while preserving the graph-level method and real-data comparison. If the Hodge summary offers little advantage over simpler measurements, that outcome will be reported honestly. Full simplicial analysis will remain deferred unless biologically meaningful higher-order structures can be justified. A synthetic-only demonstration is useful progress but does not complete the real-data milestone.

Exact dataset accessions, exposure scope, graph representation, support filters, normalization settings, numerical library and tolerances, statistical settings, and final CLI details will be recorded as implementation decisions. They do not reopen the approved research question or replace its mathematical method.